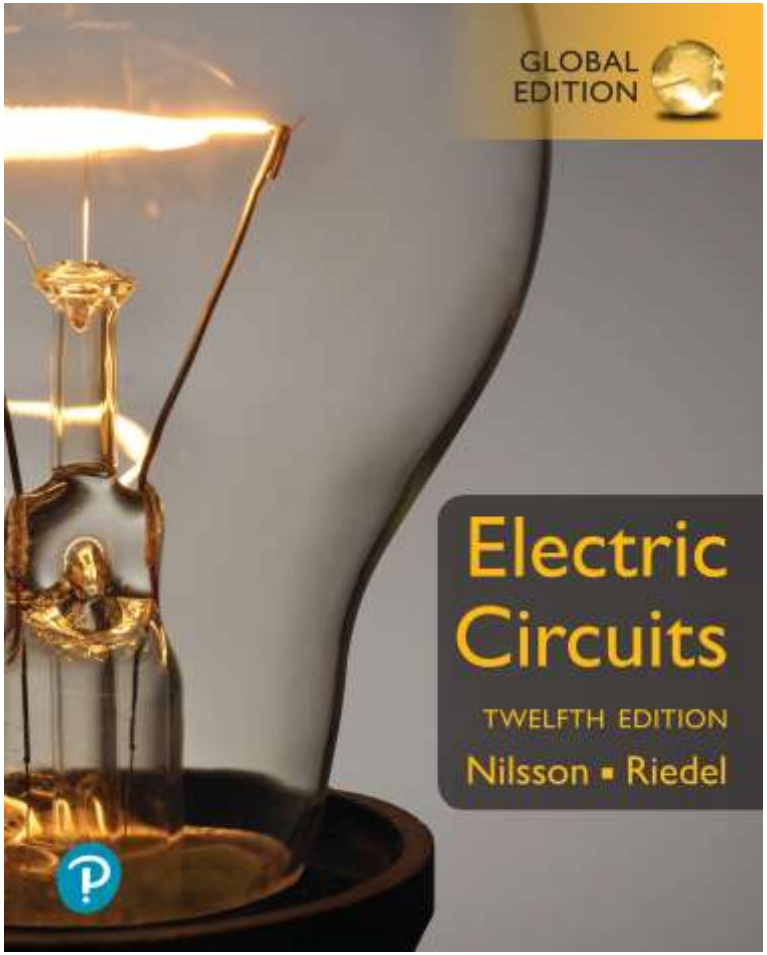
Electric Circuits

Twelfth Edition, Global Edition



Chapter 5 – Part 2

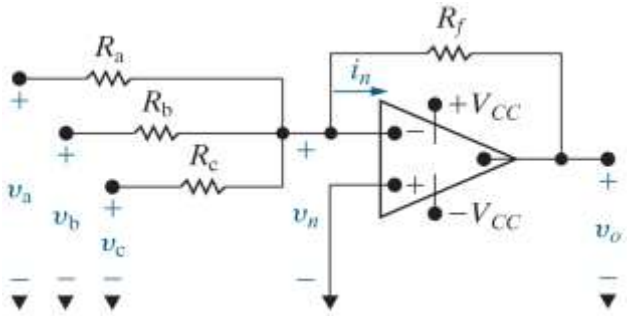
The Operational Amplifier

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Chapter 5 – Op Amps (1 of 11)

Summing Amplifier



There is a negative feedback path, so assume linear operating region (Step 1). Using the ideal op amp assumptions, write a KCL equation at the non-inverting input (Step 2) and solve it (Step 3).

$$v_p = 0 \quad \Rightarrow \quad v_n = 0$$

$$\text{KCL at } v_n: \frac{0 - v_a}{R_a} + \frac{0 - v_b}{R_b} + \frac{0 - v_c}{R_c} + \frac{0 - v_o}{R_f} = 0$$

$$\therefore \quad v_o = -\frac{R_f}{R_a} v_a - \frac{R_f}{R_b} v_b - \frac{R_f}{R_c} v_c \quad \text{for } -V_{CC} \le v_o \le V_{CC}$$

$$\text{If } R_a = R_b = R_c = R_s \quad \therefore \quad v_o = -\frac{R_f}{R_s}(v_a + v_b + v_c)$$

$$\text{If } R_a = R_b = R_c = R_s = R_f \quad \therefore \quad v_o = -(v_a + v_b + v_c)$$

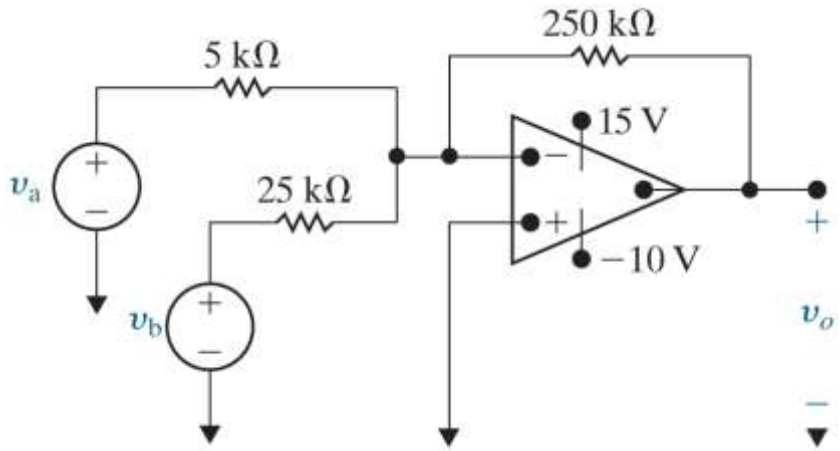
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Chapter 5 – Op Amps (2 of 11)

Summing Amplifier Example

Find v_o if $v_a = 0.1\,\text{V}$ and $v_b = 0.25\,\text{V}$.



$$v_o = -\frac{250{,}000}{5000}v_a - \frac{250{,}000}{25{,}000}v_b = -50v_a - 10v_b$$

$$\therefore \quad v_o = -50(0.1) - 10(0.25) = -7.5\ V$$

Is $-10\ V \le v_o \le 15\ V$? *Yes!*

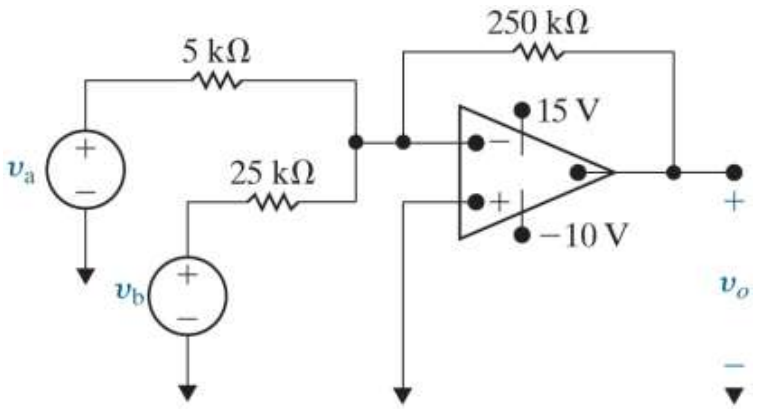
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Chapter 5 – Op Amps (3 of 11)

Summing Amplifier Example, continued

If $v_b = 0.25$ V, what positive value of v_a causes the op amp to saturate?



$$v_o = -50v_a - 10(0.25) = -50v_a - 2.5 = -10\ V$$

$$\therefore \qquad v_a = \frac{-10 + 2.5}{-50} = 0.15\ V$$

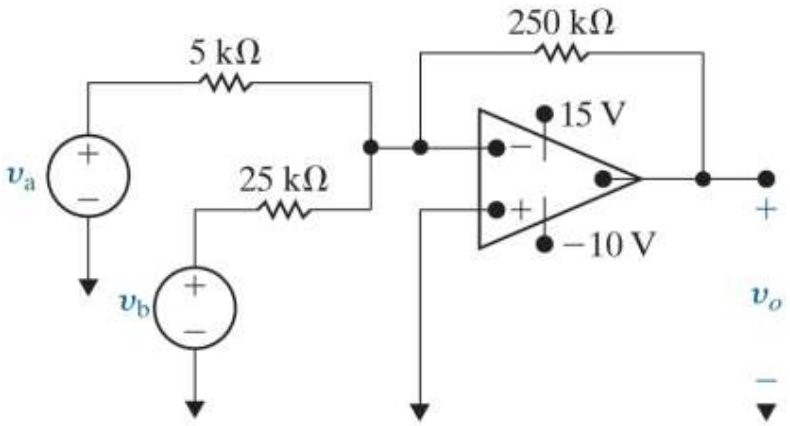
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Chapter 5 – Op Amps (4 of 11)

Summing Amplifier Example, continued

If the $250\ \text{k}\Omega$ feedback resistor is replaced with a variable resistor, how large can it be before the op amp saturates? Assume $v_a = 0.1\ \text{V}$ and $v_b = 0.25\ \text{V}$.



$$v_o = -\frac{R_f}{5000}(0.1) - \frac{R_f}{25{,}000}(0.25) = \frac{-5R_f(0.1) - R_f(0.25)}{25{,}000}$$

$$= -\frac{0.75R_f}{25{,}000} = -10\ \text{V}$$

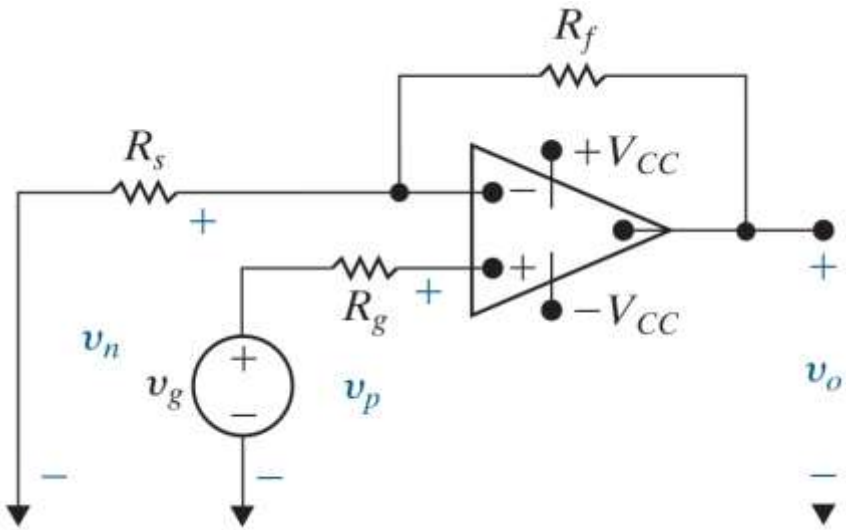
$$\Rightarrow \qquad R_f = \frac{-10(25{,}000)}{-0.75} = 333.33\ \text{k}\Omega$$

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Chapter 5 – Op Amps (5 of 11)

Non-inverting Amplifier



There is a negative feedback path, so assume linear operating region (Step 1). Using the ideal op amp assumptions, write a KCL equation at the non-inverting input (Step 2) and solve it (Step 3).

$$v_p = v_g = v_n \qquad KCL\ at\ v_n: \frac{v_g}{R_s} + \frac{v_g - v_o}{R_f} = 0$$

$$\Rightarrow \qquad v_g\left(\frac{1}{R_s} + \frac{1}{R_f}\right) = \frac{v_o}{R_f} \qquad \therefore \qquad v_o = v_g\left(\frac{R_f}{R_s} + 1\right)$$

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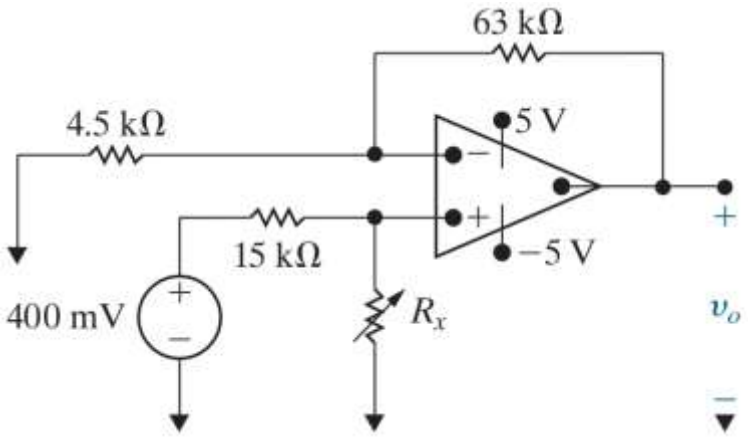
Chapter 5 – Op Amps (6 of 11)

Non-inverting Amplifier Example

Assume $R_x = 60\ \text{k}\Omega$. Find v_o.

Use voltage division to find v_p:

$$v_p = \frac{60{,}000}{15{,}000 + 60{,}000}(0.4) = 320\ \text{mV}$$



Now use KCL at the inverting input to find v_o:

$$\frac{0.32}{4500} + \frac{0.32 - v_o}{63{,}000} = 0$$

$$\Rightarrow \qquad v_o = 4.8\ \text{V}$$

Is $-5\ \text{V} \le v_o \le 5\ \text{V}$? *Yes!*

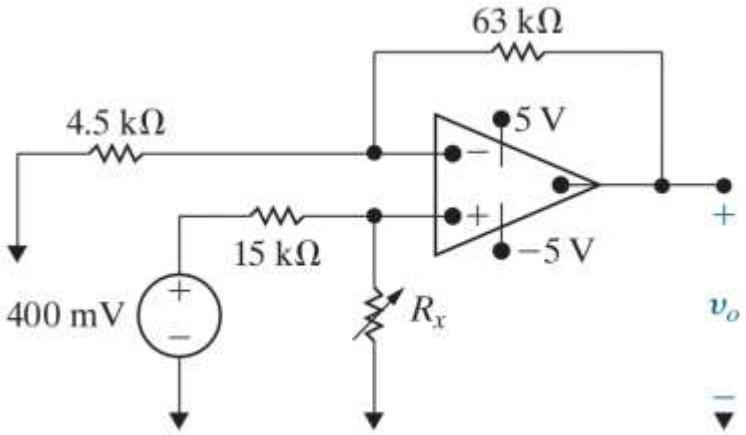
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Chapter 5 – Op Amps (7 of 11)

Non-inverting Amplifier Example, continued

Find the value of R_x that will cause the op amp to saturate.



$$v_o = \left(\frac{R_f}{R_s} + 1\right) v_p = \left(\frac{63{,}000}{4500} + 1\right) v_g = 15 v_p$$

$$v_p = \frac{R_x}{R_x + 15{,}000}(0.4) \quad \Rightarrow \quad v_o = 15\frac{R_x}{R_x + 15{,}000}(0.4) = 5$$

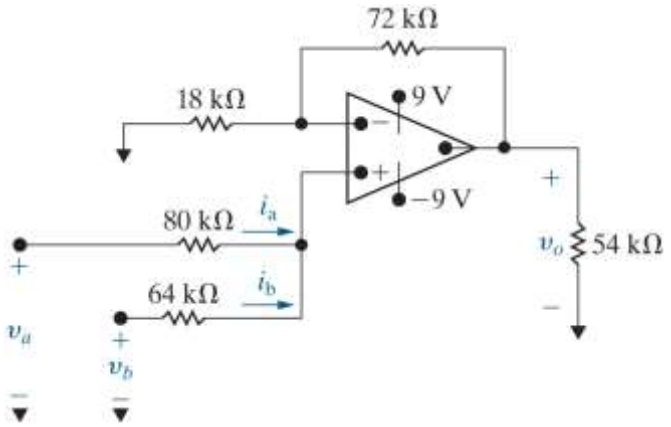
$$\Rightarrow \quad 6R_x = 75{,}000 + 5R_x \quad \therefore \quad R_x = 75\ \text{k}\Omega$$

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Chapter 5 – Op Amps (8 of 11)

What op amp circuit configuration is shown here? (Hints – how many inputs are there, and where are those inputs connected to the op amp?)



A. Inverting amplifier

B. Inverting summing amplifier

C. Non-inverting amplifier

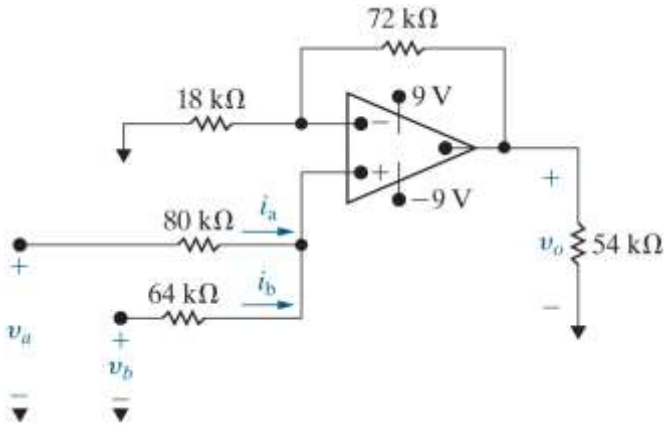
D. Non-inverting summing amplifier

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Chapter 5 – Op Amps (8 of 11)

What op amp circuit configuration is shown here? (Hints – how many inputs are there, and where are those inputs connected to the op amp?)



✘ A. Inverting amplifier

✘ B. Inverting summing amplifier

✘ C. Non-inverting amplifier

✔ D. Non-inverting summing amplifier

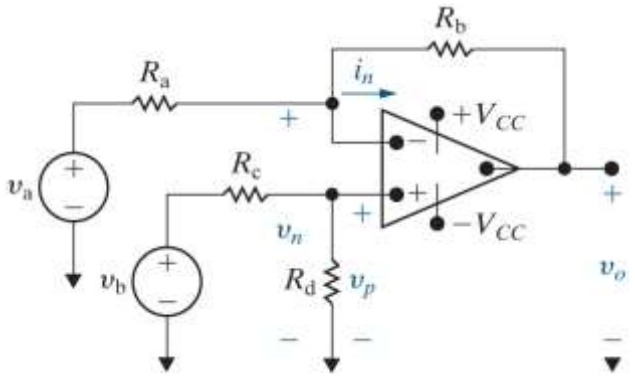
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Chapter 5 – Op Amps (9 of 11)

Difference amplifier

Voltage division: $v_p = \dfrac{R_d}{R_c + R_d} v_b$



$$\text{KCL at } v_n: \quad \frac{v_n - v_a}{R_a} + \frac{v_n - v_o}{R_b} = 0$$

$$\Rightarrow \qquad v_n(R_a + R_b) - R_b v_a = R_a v_o$$

$$\Rightarrow \qquad v_o = v_n \frac{(R_a + R_b)}{R_a} - v_a \frac{R_b}{R_a} = \frac{R_d}{R_a} \frac{(R_a + R_b)}{(R_c + R_d)} v_b - v_a \frac{R_b}{R_a}$$

If $\dfrac{R_a}{R_b} = \dfrac{R_c}{R_d}$ $\Rightarrow$ $v_o = \dfrac{R_a R_d + R_b R_d}{R_a R_c + R_a R_d} v_b - v_a \dfrac{R_b}{R_a}$

$$= \frac{R_b R_c + R_b R_d}{R_a R_c + R_a R_d} v_b - v_a \frac{R_b}{R_a} = \frac{R_b}{R_a}(v_b - v_a)$$

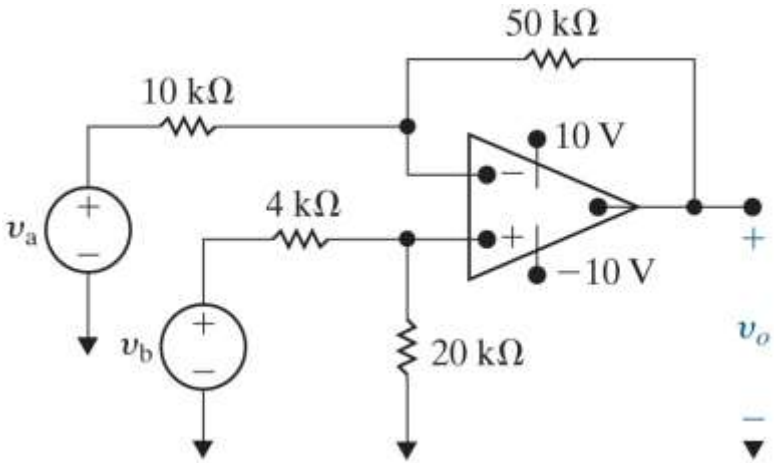
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Chapter 5 – Op Amps (10 of 11)

In a difference amplifier, if $R_a / R_b = R_c / R_d$, we can use the simple expression $v_o = R_b/R_a(v_b - v_a)$.

Can we use this simple expression for the circuit shown?



✔ A. Yes

✘ B. No

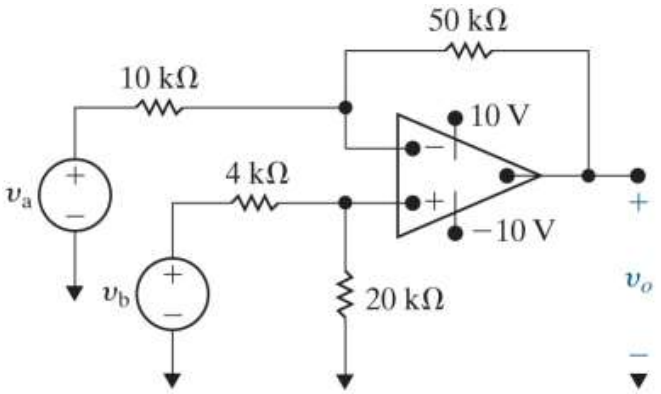
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Chapter 5 – Op Amps (11 of 11)

Difference amplifier Example

Suppose $v_b = 4$ V. What range of values for v_a will assure that the op amp operates within its linear region?



Use the simplified equation:

$$v_o = \frac{20{,}000}{4000}(v_b - v_a) = 5(v_b - v_a)$$

$$5(4 - v_a) = 10 \quad \Rightarrow \quad v_a = 2 \text{ V}$$

$$5(4 - v_a) = -10 \quad \Rightarrow \quad v_a = 6 \text{ V}$$

$$\therefore \quad 2 \text{ V} \le v_a \le 6 \text{ V}$$

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